

Placement progress and notes

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1 Monthly Progress

Month 1: 23/02/2026 - 20/03/2026

Month 2: 23/03/2026 - 24/04/2026

Overview

Completed dynamic aperture and momentum acceptance studies for different configurations. Experimented with slicing, integrators, and fringe fields. Now introducing misalignments and corrections.

Work completed

- Plotted DA/MA for 7 different sextupole configurations
- Optimised configurations to increase DA/MA
- Changed number of slices in integrators to increase quality of plots
- Introduced fringe fields for every magnet type and compared with "perfect" results
- Performed frequency map analysis
- Introduced misalignments, inserted BPMs and correctors, and performed orbit correction

Problems encountered

Reducing non-linearities: Chromatic correction is working for dqx and dqy but often plotting tune against momentum deviation showed that $ddqx$ and $ddqy$ had not been properly corrected. This contributed to smaller DA/MA. Performing chromaticity correction to get these values to 0 often failed as it was too much of a jump.

- **Solution:** The chromaticity correction tries to match to 80% of the value of $ddqx$ and $ddqy$. If this is successful, it iterates through 80% of the resulting values until the code fails. It takes the strengths from the iteration with the smallest second order chromaticity.

Code organisation: Many scripts and results in different places

- **Solution:** Everything has been modularised so it is easier to see where errors arise and can edit things more efficiently. This will also help with later lattice designs.

Results and progress

- Increasing the number of slices for the quadrupoles created smoother curves in the DA/MA plots, but there were no differences when increasing the slices for bends and quadrupoles.
- Fringe fields reduced the less stable configurations' DA/MA considerably, but not as much for stable configurations.
- Added another focusing sextupole pair to the configuration with two sextupole pairs at 180° phase advance, which dramatically improved momentum acceptance.
- Frequency map analysis calculated and shows stable diffusion
- BPMs and correctors are currently placed around every quadrupole (horizontal and vertical) and the MICADO algorithm is used for corrections
- With corrections, the orbits are currently oscillating at less than 1mm

Next steps

- Studies on spin tracking and polarization build-up time
- Looking at integration with the rest of the injector complex using current macroparticle distribution
- Assess injection efficiency

Skills developed

- **Dynamic aperture and momentum acceptance:** how it is calculated and the effects of adjusting integrators and slices
- **Knowledge of beam optics concepts:** fringe fields, FMA, BPMs, correctors, synchrotron radiation damping

Glossary

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